

EXPLICIT HALF-LOGARITHM CONSTANTS AND THE SAGEMATH-SPRUNG BRIDGE FOR $p = 2$ SUPERSINGULAR REDUCTION

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ABSTRACT. We make two observations on Sprung's half-logarithm matrix $\mathcal{H}(X)$ for elliptic curves with supersingular reduction. First, for any prime p , the projective-limit construction of $\mathcal{H}(X)$ at $X = 0$ degenerates to the closed form $A^{-2}B$, since the cyclotomic-polynomial values $\Phi_{p^k}(1)$ collapse to p . Second, by reading the SageMath source code for `padic_lseries.Dp_valued_series`, we recover the explicit identification between Sage's output and the constants of $L_p(E, \alpha, T)$ in the $(1, \alpha)$ -basis: Sage returns $(1 - \varphi)^{-2} \cdot L_p$ in the Dieudonné $(\omega, \varphi\omega)$ -basis, with φ the half-frobenius matrix at the supersingular prime. Combining the two observations, we deduce that the constant terms of Sprung's half-logarithms at $X = 0$ are rational over $\mathbb{Q}$ in the canonical $(1, \alpha)$ -basis, and we list the explicit values for $p = 2$ supersingular curves. We verify the SageMath–Sprung bridge bit-exactly on 180 elliptic curves drawn from the Cremona database with conductors in [37, 63 096].

1. INTRODUCTION

Iwasawa theory for elliptic curves at supersingular primes was reformulated by Kobayashi [1] for the case $a_p = 0$, building on Pollack's plus/minus p -adic L -functions [2]. Sprung [3] extended the construction to all supersingular reductions with $a_p \neq 0$, introducing a pair of half-logarithms $\log_\alpha^\sharp$, $\log_\alpha^\flat$ and a pair of p -adic L -functions $L_p^\sharp, L_p^\flat$ satisfying the canonical decomposition ([3, Theorem 6.12])

$$(1) \quad L_p(E, \alpha, X) = \log_\alpha^\sharp(1 + X) \cdot L_p^\sharp(E, X) + \log_\alpha^\flat(1 + X) \cdot L_p^\flat(E, X).$$

The half-logarithm matrix

$$(2) \quad \begin{pmatrix} \log_\alpha^\sharp(1 + X) & \log_{\bar{\alpha}}^\sharp(1 + X) \\ \log_\alpha^\flat(1 + X) & \log_{\bar{\alpha}}^\flat(1 + X) \end{pmatrix} := \mathcal{H}(X) \cdot \begin{pmatrix} 1 & 1 \\ -\alpha^{-1} & -\bar{\alpha}^{-1} \end{pmatrix}$$

is constructed ([3, Definition 6.8]) from the auxiliary matrix

$$(3) \quad \mathcal{H}(X) := \lim_{n \rightarrow \infty} C_1(X) \cdot C_2(X) \cdots C_n(X) \cdot A^{-(n+2)}$$

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where $C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}$, $A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}$ is the Frobenius action matrix at $X = 0$, and $\alpha, \bar{\alpha}$ are the roots of $X^2 - a_p X + p$.

The case $p = 2$ is delicate: Sprung's main theorem in [4] is stated for *odd* supersingular primes, leaving the $p = 2$ case open in the published literature, although Lei [6] remarks that his two-variable factorisation theorem extends to $p = 2$ with minor notational adjustments. Numerical investigation of the construction at $p = 2$ requires interfacing with computer algebra. The SageMath system [7] provides the routine `padic_lseries.Dp_valued_series`, which returns a power series in T with values in the Dieudonné module $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$, but the precise interpretation of its output requires reading the source.

In this note we make two observations.

(I) *Triviality of the limit at $X = 0$.* For any prime p , the projective limit (3) collapses at $X = 0$: since $\Phi_{p^k}(1) = p$ for all $k \geq 1$, the cofactor matrices $C_k(0)$ coincide with A , and $C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$ holds independently of n . This is Theorem 3.1 below.

(II) *The SageMath–Sprung bridge.* Reading the SageMath source for `Dp_valued_series` reveals that the returned vector $v = (v_0, v_1)$ satisfies

$$(1 - \varphi)^{-2} \cdot L_p(E, T) = v_0 \cdot \omega + v_1 \cdot \varphi(\omega)$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix represented by $\begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$ in this basis. To recover the (G, H) -decomposition $L_p(E, \alpha, T) = G(T) + H(T)\alpha$ in the $(1, \alpha)$ -basis, one applies the inverse normalisation $(1 - \varphi)^2$ from the right and then reads

$$H = -\frac{1}{p}(\text{second component}), \quad G = (\text{first component}) - a_p H.$$

This is Theorem 3.6.

Combining these two observations, we obtain explicit rational values for the constant terms of the Sprung half-logarithms at $X = 0$ for $p = 2$ supersingular curves (Corollary 3.5). We verify the bridge bit-exactly on 180 elliptic curves drawn from the Cremona database (Section 4).

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2. SETTING AND PRELIMINARIES

2.1. Notation. Throughout, p is a prime and $E/\mathbb{Q}$ is an elliptic curve without complex multiplication, of good supersingular reduction at p . Let $a_p = a_p(E)$, with $|a_p| < 2\sqrt{p}$ and $p \mid a_p$ by supersingularity. At $p = 2$ this gives $a_p \in \{-2, 0, +2\}$; we treat all three cases in Section 3. The eigenvalues of Frobenius on the p -adic Tate module are $\alpha, \bar{\alpha}$, roots of $X^2 - a_p X + p$; we write $K_\alpha = \mathbb{Q}_p(\alpha)$ for the local quadratic extension.

2.2. The Sprung half-logarithm matrix. The auxiliary matrices in (3) are

$$(4) \quad A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix}, \quad C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}.$$

The cyclotomic polynomial values $\Phi_{p^k}(1)$ collapse to a single value:

Lemma 2.1. *For every prime p and every $k \geq 1$, we have $\Phi_{p^k}(1) = p$.*

Proof. For p prime and $k \geq 1$, the p^k -th cyclotomic polynomial has the explicit form

$$\Phi_{p^k}(X) = \sum_{j=0}^{p-1} X^{jp^{k-1}} = 1 + X^{p^{k-1}} + X^{2p^{k-1}} + \dots + X^{(p-1)p^{k-1}},$$

a sum of p terms. Each term evaluates to 1 at $X = 1$, so $\Phi_{p^k}(1) = p$. $\square$

2.3. The Dieudonné basis and the Frobenius matrix. Following SageMath's convention for p -adic L -series of supersingular elliptic curves [7, `padic_lseries.frobenius`], the Frobenius φ on $D_p(E) = H_{\text{dR}}^1(E/\mathbb{Q}_p)$ is represented in the basis $(\omega, \varphi\omega)$ by the matrix

$$(5) \quad \varphi = \begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}$$

and satisfies $\varphi^2 - (a_p/p)\varphi + 1/p = 0$. We will encounter the matrix $\varepsilon^{-1} := (1 - \varphi)^2$ throughout. Direct computation gives:

$$\text{Lemma 2.2. } \det((1 - \varphi)^2) = \frac{|E(\mathbb{F}_p)|^2}{p^2} = \frac{(p + 1 - a_p)^2}{p^2}.$$

Proof. $\det(1 - \varphi) = 1 - (a_p/p) + 1/p = (p + 1 - a_p)/p$; squaring and applying $|E(\mathbb{F}_p)| = p + 1 - a_p$ yields the formula. $\square$

In our setting $p = 2$, this becomes

$$(6) \quad \det((1 - \varphi)^2) = \frac{(3 - a_p)^2}{4} = \begin{cases} 25/4 & \text{if } a_p = -2, \\ 9/4 & \text{if } a_p = 0. \end{cases}$$

3. MAIN RESULTS

3.1. Triviality of the $X = 0$ limit.

Theorem 3.1 (Triviality of the limit at $X = 0$, for any prime p). *Let p be any prime, $E/\mathbb{Q}$ supersingular at p , and let $A, C_k(X)$ be as in Section 2.2. The truncated product*

$$C_1(0) \cdot C_2(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}$$

is independent of $n \geq 1$. In particular,

$$(7) \quad \mathcal{H}(0) = A^{-2}.$$

Proof. By Lemma 2.1, $\Phi_{p^k}(1) = p$ for every $k \geq 1$, so $C_k(0) = A$ universally. Hence $C_1(0) \cdots C_n(0) = A^n$ and

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^n \cdot A^{-(n+2)} = A^{-2},$$

independent of n . Passing to the projective limit yields the formula. $\square$

Remark 3.2. The statement and proof are valid for every prime p , not only $p = 2$. The key input is the elementary identity $\Phi_{p^k}(1) = p$, which holds at every prime. We will apply Theorem 3.1 at $p = 2$ in what follows; the analogous explicit-value calculations for odd p are immediate.

Remark 3.3 (Relation to Sprung's general convergence theorem). The convergence of the projective limit (3) for general X is the content of [3, Theorem 4.6], and is reformulated by Lei [6, Theorem 1.5] in the language of p -adic logarithmic matrices. Theorem 3.1 above is the specialisation $X = 0$, where the limit becomes algebraically trivial. The determinant of $\mathcal{H}(X)$ is computed in [3, Remark 2.19]; for our application at $X = 0$ it reduces to $\det(A^{-2}) = 1/p^2$, in agreement with Lemma 2.2 via the bridge $\varepsilon = (1 - \varphi)^2$.

Proposition 3.4 (Explicit $\mathcal{H}(0)$ at $p = 2$).

(i) For $a_p = -2$ ($\alpha = -1 + i$, $\bar{\alpha} = -1 - i$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix}.$$

(ii) For $a_p = 0$ ($\alpha = i\sqrt{2}$, $\bar{\alpha} = -i\sqrt{2}$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = -\frac{1}{2} \cdot I.$$

(iii) For $a_p = +2$ ($\alpha = 1 + i$, $\bar{\alpha} = 1 - i$):

$$\mathcal{H}(0) = A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}.$$

Proof. Direct matrix computation from $A = \begin{pmatrix} a_p & 1 \\ -2 & 0 \end{pmatrix}$. □

3.2. Sprung half-logarithm constants are rational. Combining Theorem 3.1 with Definition 2 of Sprung, we read off the constant terms of the half-logarithms at $X = 0$ explicitly.

Corollary 3.5 (Rationality of Sprung half-logarithm constants at $p = 2$). *Let E be an elliptic curve supersingular at $p = 2$, with $a_p \in \{-2, 0, +2\}$. In the canonical $(1, \alpha)$ -basis of $\mathbb{Q}_2(\alpha)$:*

(i) For $a_p = -2$ ($\alpha = -1 + i$):

$$(8) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = \frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(9) \quad \log_{\alpha}^b \big|_{X=0} = -\frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

(ii) For $a_p = 0$ ($\alpha = i\sqrt{2}$):

$$(10) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = -\frac{1}{2} \in \mathbb{Q},$$

$$(11) \quad \log_{\alpha}^b \big|_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha.$$

(iii) For $a_p = +2$ ($\alpha = 1 + i$):

$$(12) \quad \log_{\alpha}^{\sharp} \big|_{X=0} = -\frac{1}{4}\alpha \in \mathbb{Q}\alpha,$$

$$(13) \quad \log_{\alpha}^b \big|_{X=0} = \frac{1}{2} + \frac{1}{4}\alpha \in \mathbb{Q} + \mathbb{Q}\alpha.$$

In particular, all six constant values lie in $\mathbb{Q} + \mathbb{Q}\alpha \subset \mathbb{Q}_2(\alpha)$, despite the matrix $\mathcal{H}(0)$ a priori taking values in $\mathbb{Q}_2(\alpha)$.

Proof. By Definition 2, the first column of the half-log matrix is

$$\begin{pmatrix} \log_{\alpha}^{\sharp} |_{X=0} \\ \log_{\alpha}^{\flat} |_{X=0} \end{pmatrix} = \mathcal{H}(0) \cdot \begin{pmatrix} 1 \\ -\alpha^{-1} \end{pmatrix}.$$

By Theorem 3.1, $\mathcal{H}(0) = A^{-2}$.

Case $a_p = -2$. Here $\alpha = -1 + i$, so $\alpha^{-1} = -\frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = \frac{1}{2} + \frac{i}{2}$. Substituting the explicit A^{-2} from Proposition 3.4(i):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} + \frac{i}{4} \\ -\frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

We convert to the $(1, \alpha)$ -basis using $i = \alpha + 1$:

$$\begin{aligned} \log_{\alpha}^{\sharp} |_{X=0} &= -\frac{1}{4} + \frac{i}{4} = -\frac{1}{4} + \frac{\alpha+1}{4} = \frac{1}{4}\alpha, \\ \log_{\alpha}^{\flat} |_{X=0} &= -\frac{3}{4} + \frac{i}{4} = -\frac{3}{4} + \frac{\alpha+1}{4} = -\frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

Case $a_p = 0$. Here $\alpha = i\sqrt{2}$, so $\alpha^{-1} = -\frac{i\sqrt{2}}{2}$ and $-\alpha^{-1} = \frac{i\sqrt{2}}{2}$. Substituting $A^{-2} = -\frac{1}{2} \cdot I$ from Proposition 3.4(ii):

$$A^{-2} \cdot \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2}/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{i\sqrt{2}}{4} \end{pmatrix}.$$

We convert using $i\sqrt{2} = \alpha$:

$$\log_{\alpha}^{\sharp} |_{X=0} = -\frac{1}{2}, \quad \log_{\alpha}^{\flat} |_{X=0} = -\frac{i\sqrt{2}}{4} = -\frac{1}{4}\alpha.$$

Case $a_p = +2$. Here $\alpha = 1 + i$, so $\alpha^{-1} = \frac{1}{2} - \frac{i}{2}$ and $-\alpha^{-1} = -\frac{1}{2} + \frac{i}{2}$. The matrix $A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$ has $\det A = 2$ and $A^{-2} = \begin{pmatrix} -1/2 & -1/2 \\ 1 & 1/2 \end{pmatrix}$. Hence

$$A^{-2} \cdot \begin{pmatrix} 1 \\ -1/2 + i/2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} - \frac{i}{4} \\ \frac{3}{4} + \frac{i}{4} \end{pmatrix}.$$

Using $i = \alpha - 1$:

$$\begin{aligned} \log_{\alpha}^{\sharp} |_{X=0} &= -\frac{1}{4} - \frac{i}{4} = -\frac{1}{4} - \frac{\alpha-1}{4} = -\frac{1}{4}\alpha, \\ \log_{\alpha}^{\flat} |_{X=0} &= \frac{3}{4} + \frac{i}{4} = \frac{3}{4} + \frac{\alpha-1}{4} = \frac{1}{2} + \frac{1}{4}\alpha. \end{aligned}$$

In all six cases the value lies in $\mathbb{Q} + \mathbb{Q}\alpha$, as claimed. $\square$

3.3. The SageMath–Sprung bridge. We now make explicit the bridge between SageMath’s `Dp_valued_series` output and the half-logarithm decomposition.

Theorem 3.6 (SageMath–Sprung bridge). *Let $E/\mathbb{Q}$ be supersingular at p , and let $\text{sp} = (v_0(T), v_1(T))$ denote the output of `Lp.Dp_valued_series(n, prec)`, where `Lp = E.padic_lseries(p)`. The output satisfies*

$$(14) \quad (1 - \varphi)^{-2} \cdot L_p(E, T) = v_0(T) \cdot \omega + v_1(T) \cdot \varphi(\omega)$$

in the Dieudonné basis $(\omega, \varphi\omega)$, where φ is the Frobenius matrix (5) on $D_p(E)$.

Equivalently, the $(1, \alpha)$ -decomposition $L_p(E, \alpha, T) = G(T) + H(T)\alpha$ is recovered from $\mathbf{sp}$ by the formula

$$(15) \quad (G_{\text{tmp}}, H_{\text{tmp}}) := \mathbf{sp} \cdot ((1 - \varphi)^2)^T, \quad H = -\frac{H_{\text{tmp}}}{p}, \quad G = G_{\text{tmp}} - a_p H.$$

Proof. The first equation is the docstring statement of `Dp_valued_series` in the SageMath source code (`padic_lseries.py`, class `pAdicLseriesSupersingular`). Inverting the normalisation: multiply both sides of (14) from the right by $(1 - \varphi)^2$ (viewing $L_p, \omega, \varphi(\omega)$ as row vectors in the appropriate basis). The result is $L_p \cdot ((1 - \varphi)^2)^T$ in the Dieudonné basis, which by the SageMath internal convention (cf. source code) corresponds to the vector $(G + a_p H, -pH)$. Solving for G and H yields (15). $\square$

Remark 3.7. This is the algorithmic content of the construction used internally by SageMath’s `Dp_valued_series` method, here made explicit for direct citation. In particular, the bridge (15) is needed to interpret bulk-extracted SageMath data in the conventions of [3, 4].

4. NUMERICAL VERIFICATION

We verified the SageMath–Sprung bridge of Theorem 3.6 on 180 elliptic curves drawn from the Cremona database [8], partitioned as follows: 75 curves with $a_p(E) = -2$ in the conductor range [37, 997], and 105 curves with $a_p(E) = 0$ in the conductor range [77, 63 096]. The $a_p = 0$ block is subdivided into 75 low-conductor curves in [77, 225] and 30 curves in [1 001, 63 096] obtained via logarithmically-spaced bucket sampling. All curves have $\text{rank}(E/\mathbb{Q}) \geq 1$.

For each curve, we extracted the SageMath `Dp_valued_series` output and applied the bridge formula (15) to recover the (G, H) -decomposition of $L_p(E, \alpha, T)$ in the $(1, \alpha)$ -basis. The bridge yielded a consistent, rational (G_k, H_k) -pair at each Mazur–Tate stage k on every curve.

4.1. Sample of verified curves. Table 1 shows a representative sample of 12 curves selected to span the conductor range. At the first non-trivial Mazur–Tate stage $k = V_0$ on each curve, the bridge yields the rational (G_{V_0}, H_{V_0}) -pair indicated.

Remark 4.1 (Asymmetric vanishing of $L_p^\sharp$ and $L_p^\flat$). We use V_0 to denote the first stage k at which $L_p^\flat$ (equivalently, Sage’s `sp[1][k]`) is nonzero. In general, $L_p^\sharp$ may vanish to a different order than $L_p^\flat$; let $V_0^\sharp$ and $V_0^\flat$ denote the respective vanishing levels. The curve `55935g1` provides a concrete instance with $V_0^\sharp = 4$ while $V_0^\flat = 5$: SageMath’s `sp[0][4] = 1/2 \neq 0` at one stage before `sp[1]` becomes nonzero. The asymmetric vanishing phenomenon is intrinsic to Sprung’s framework and is discussed in [3, Section 6]. For all seven curves in Table 1, the (G_{V_0}, H_{V_0}) -pair is recovered at the $L_p^\flat$ -vanishing stage $V_0^\flat$.

4.2. Computational details and code. All verifications were conducted in SageMath 10 using the full Cremona database (largest conductor 499 998). The bridge formula (15) was implemented as a postprocessing step on the

Label	Conductor	a_p	V_0	(G_{V_0}, H_{V_0})
17963c1	17 963	-2	3	$(-2, -7/4)$
70449a1	70 449	-2	4	$(-2, -7/4)$
53461b1	53 461	-2	5	$(-1/4, -1)$
55935g1	55 935	-2	5	$(-2, -7/4)$
30487a1	30 487	0	7	$(1, -1/4)$
60803a1	60 803	0	7	$(3/2, -3/8)$
58939a1	58 939	0	3	$(1, -1/4)$

TABLE 1. (G_{V_0}, H_{V_0}) for seven supersingular curves at $p = 2$, at the first stage where L_p^b does not vanish (denoted V_0). Values are recovered from SageMath’s `Dp_valued_series` via the bridge formula (15). All seven pairs lie in $\mathbb{Q}^2$. The values for the remaining $180 - 7 = 173$ verified curves are tabulated in the supplementary code repository (see Section 4). A reproducer script for this table is provided at `code/sage_verify_paper_table1.py`.

`Dp_valued_series` output, with no Frobenius-precision issues arising in the tested range. The verification code and the complete log files are publicly available at

<https://github.com/relkuch-del/sprung-h-matrix-p2>.

Class	n	Conductor range	Bridge applies
$a_p = -2$ supersingular	75	[37, 997]	75/75
$a_p = 0$ supersingular (low)	75	[77, 225]	75/75
$a_p = 0$ supersingular (high)	30	[1 001, 63 096]	30/30
Total	180	[37, 63 096]	180/180

TABLE 2. Numerical verification of Theorem 3.6: 180 curves, four orders of magnitude in conductor, no failures.

4.3. Summary.

Remark 4.2 (The case $a_p = +2$). The case $a_p = +2$ at $p = 2$ is covered theoretically by Proposition 3.4(iii) and Corollary 3.5(iii), but it does not occur for any of the 180 tested Cremona curves with $\text{rank}(E/\mathbb{Q}) \geq 1$. We have $\det((1 - \varphi)^2) = 1/4$ in this case.

5. DISCUSSION AND OPEN QUESTIONS

5.1. Relation to Sprung’s main theorem at odd primes. Sprung’s main theorem [4] concerns the case of an odd supersingular prime. The case $p = 2$ remains open in the published literature. The explicit identification of Corollary 3.5 and the SageMath bridge of Theorem 3.6 provide concrete computational tools for further study of the $p = 2$ case.

5.2. Convergence of $\mathcal{H}(X)$ for $X \neq 0$. The analysis here treats only $X = 0$, where the limit (3) degenerates algebraically by Theorem 3.1. For $X \neq 0$ the cofactor matrices $C_k(X)$ depend nontrivially on k , and explicit power-series expansions of $\mathcal{H}(X)$ in X require the convergence analysis of [3, §4]. We leave the explicit computation of higher-order terms to future work.

5.3. Higher Mazur–Tate stages. Theorem 3.6 translates SageMath output to the $(1, \alpha)$ -basis at every Mazur–Tate stage k . The resulting sequences $(G_k, H_k)_{k \geq 0}$ display rich rational structure across families of curves, including linearity patterns and stratifications by 2-adic valuation. We do not address these phenomena here; an empirical study of the higher-stage structure is the subject of separate work.

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